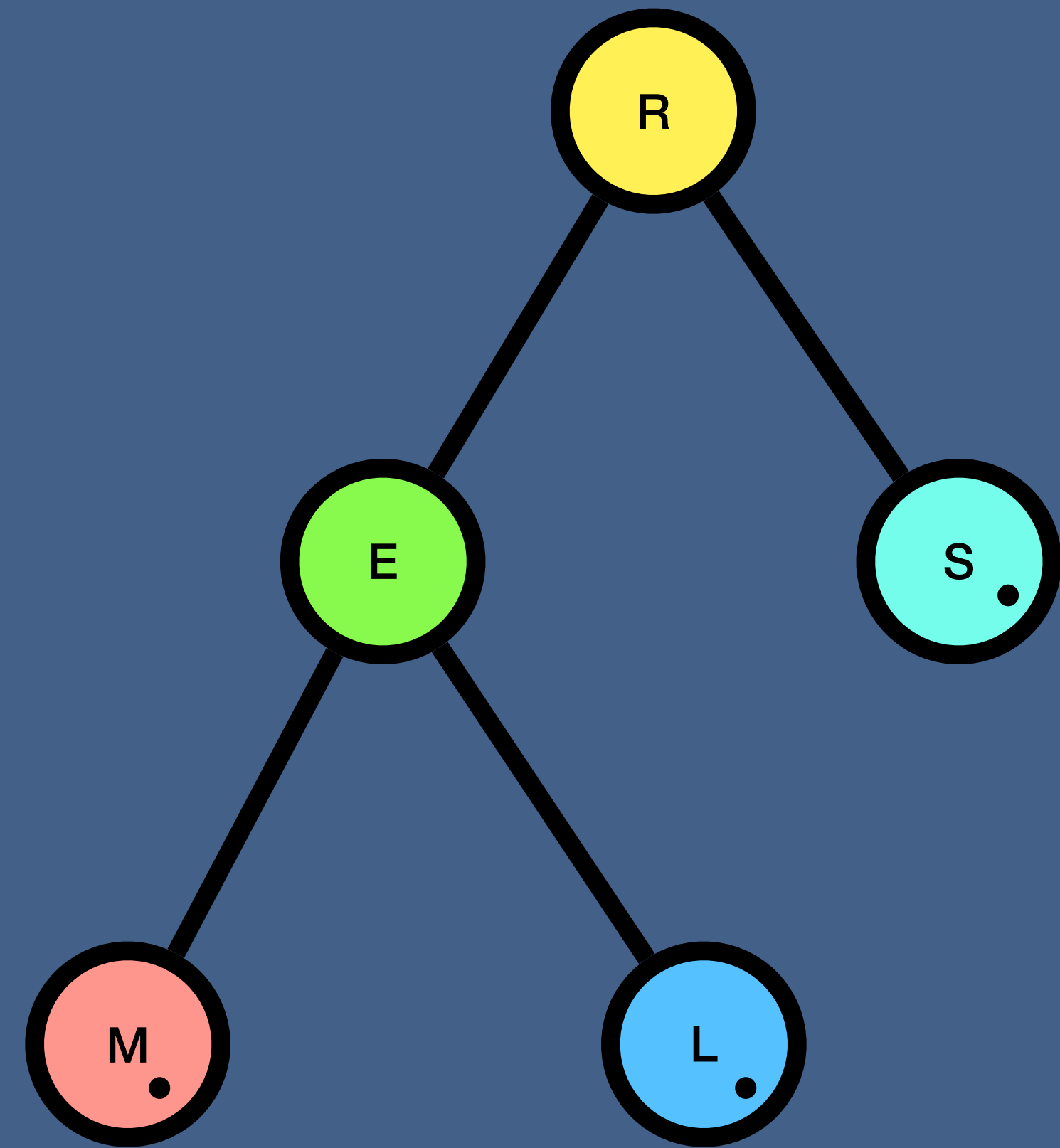


Trie



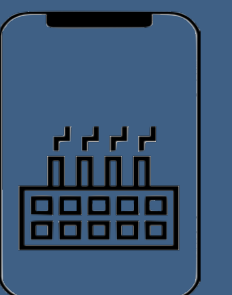
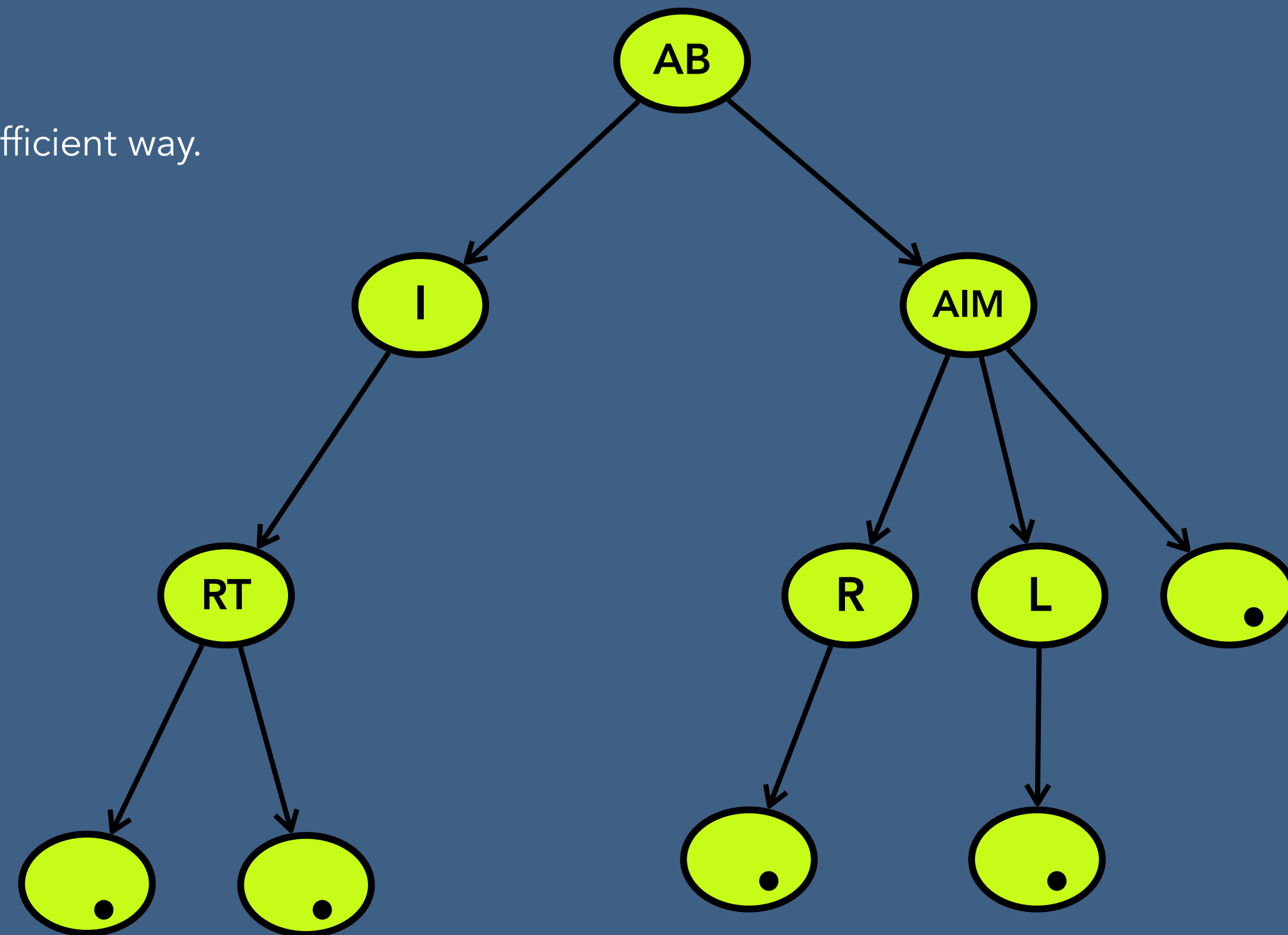
What is a Trie?

A Trie is a tree-based data structure that organizes information in a hierarchy.

Properties:

- It is typically used to store or search strings in a space and time efficient way.
- Any node in trie can store non repetitive multiple characters
- Every node stores link of the next character of the string
- Every node keeps track of 'end of string'

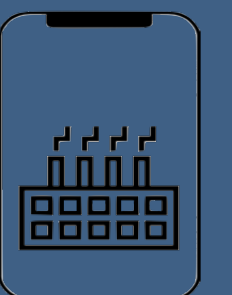
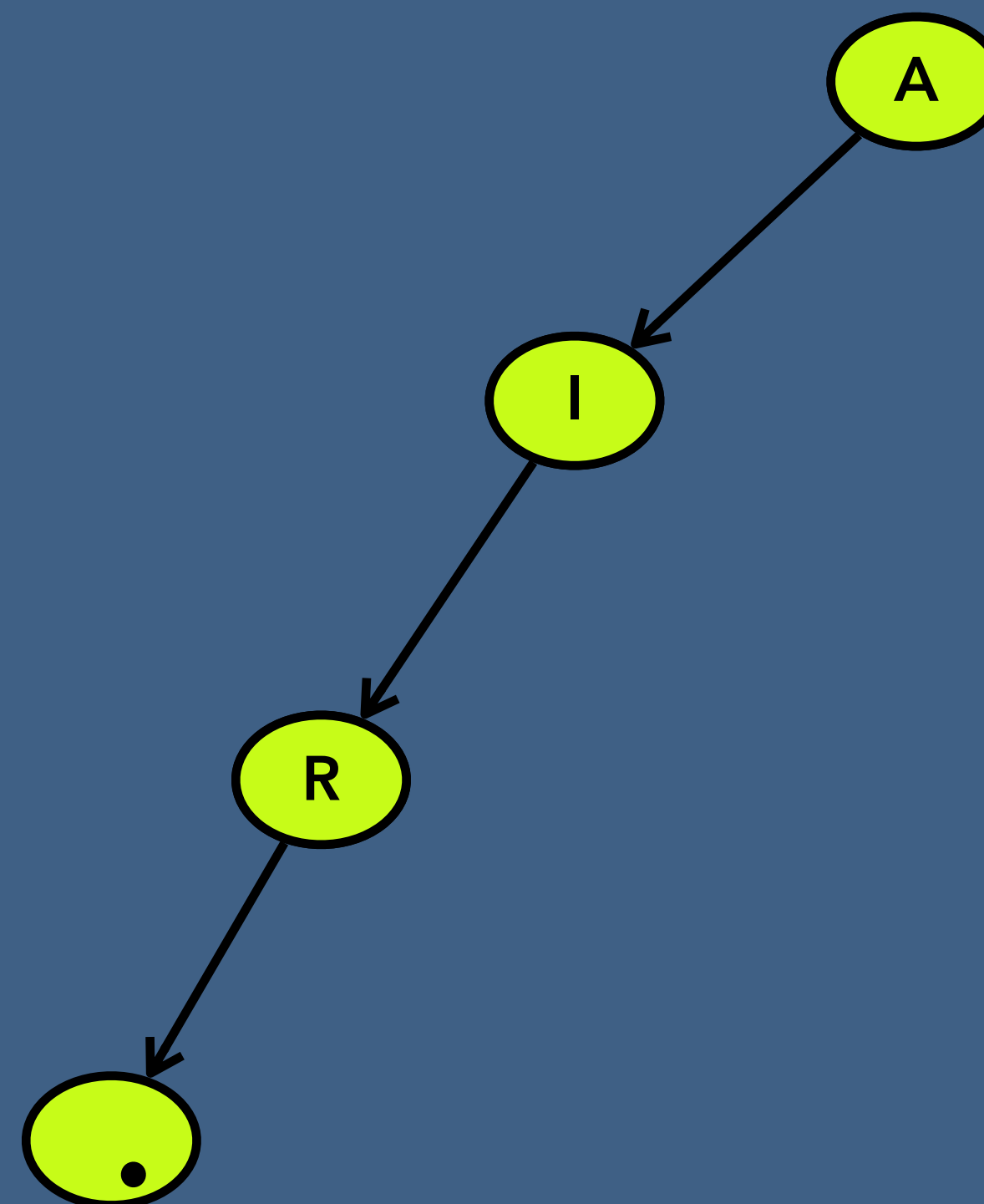
AIR, AIT, BAR, BIL, BM



What is a Trie?

A Trie is a tree-based data structure that organizes information in a hierarchy.

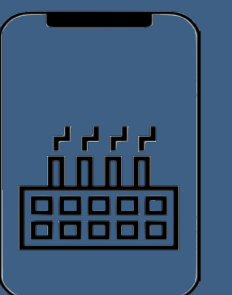
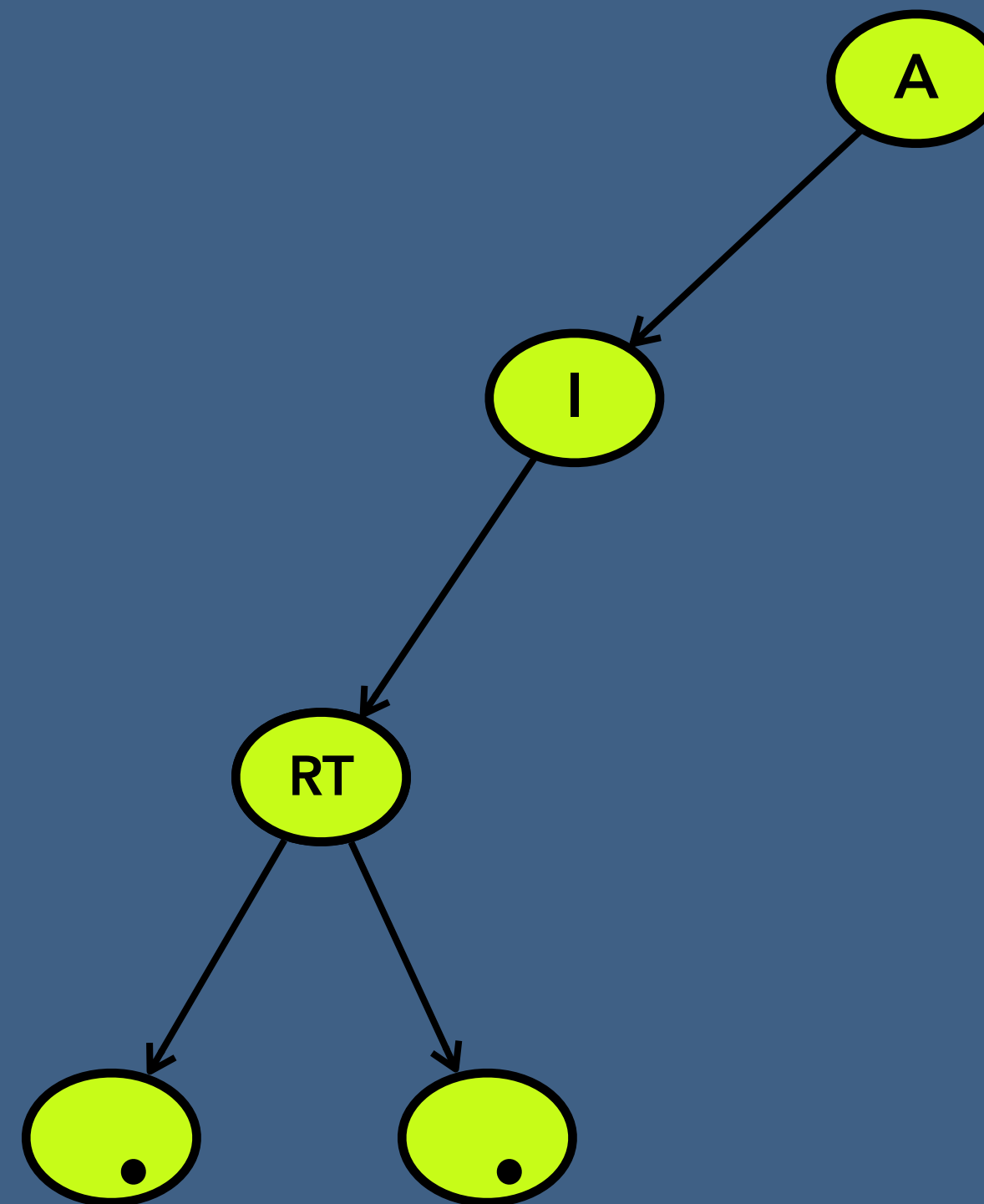
AIR



What is a Trie?

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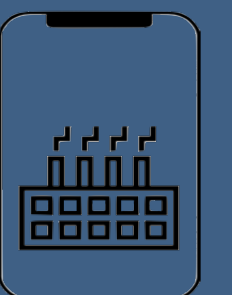
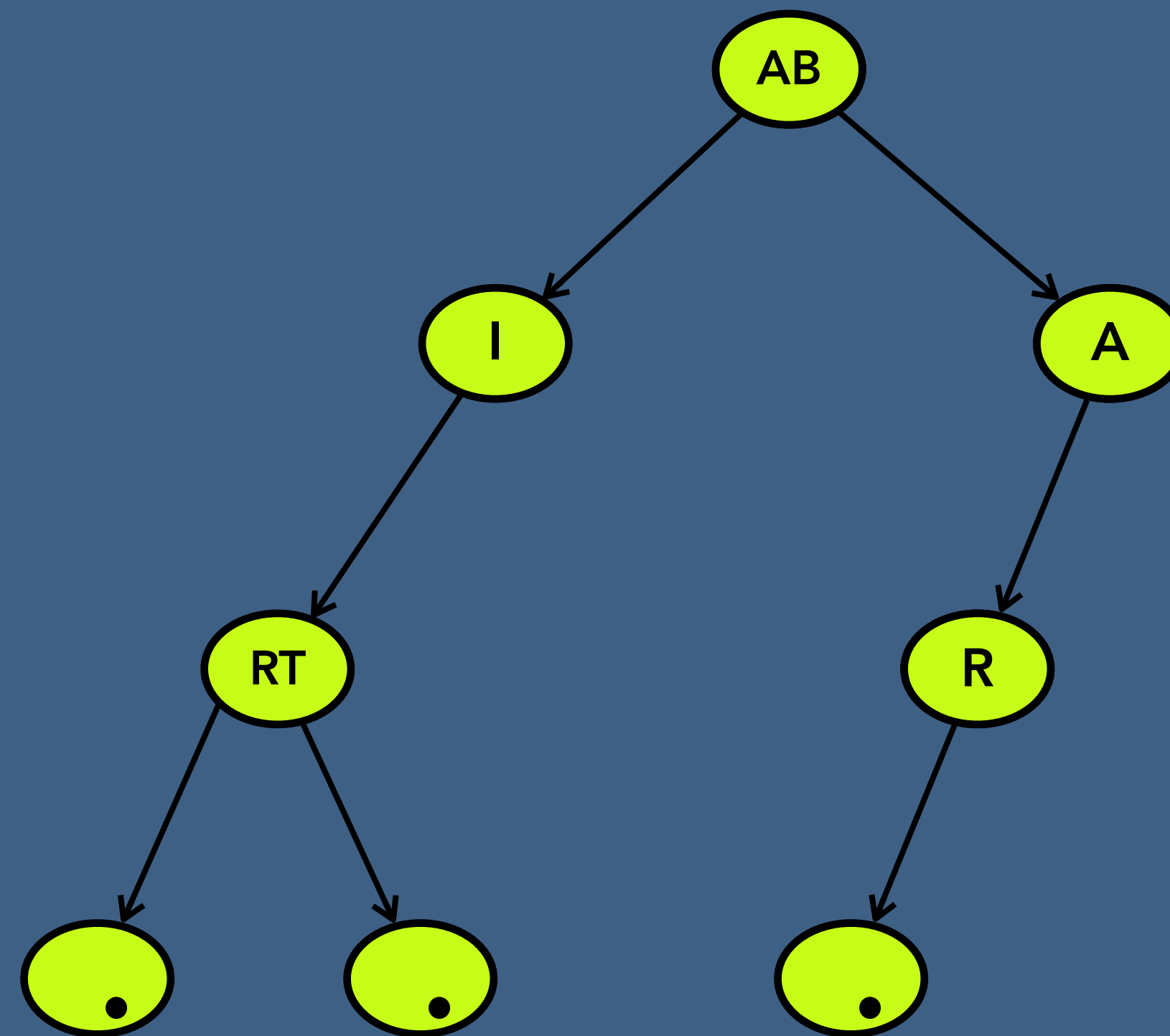
AIT



What is a Trie?

A Trie is a tree-based data structure that organizes information in a hierarchy.

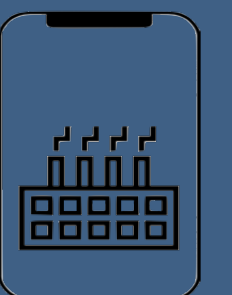
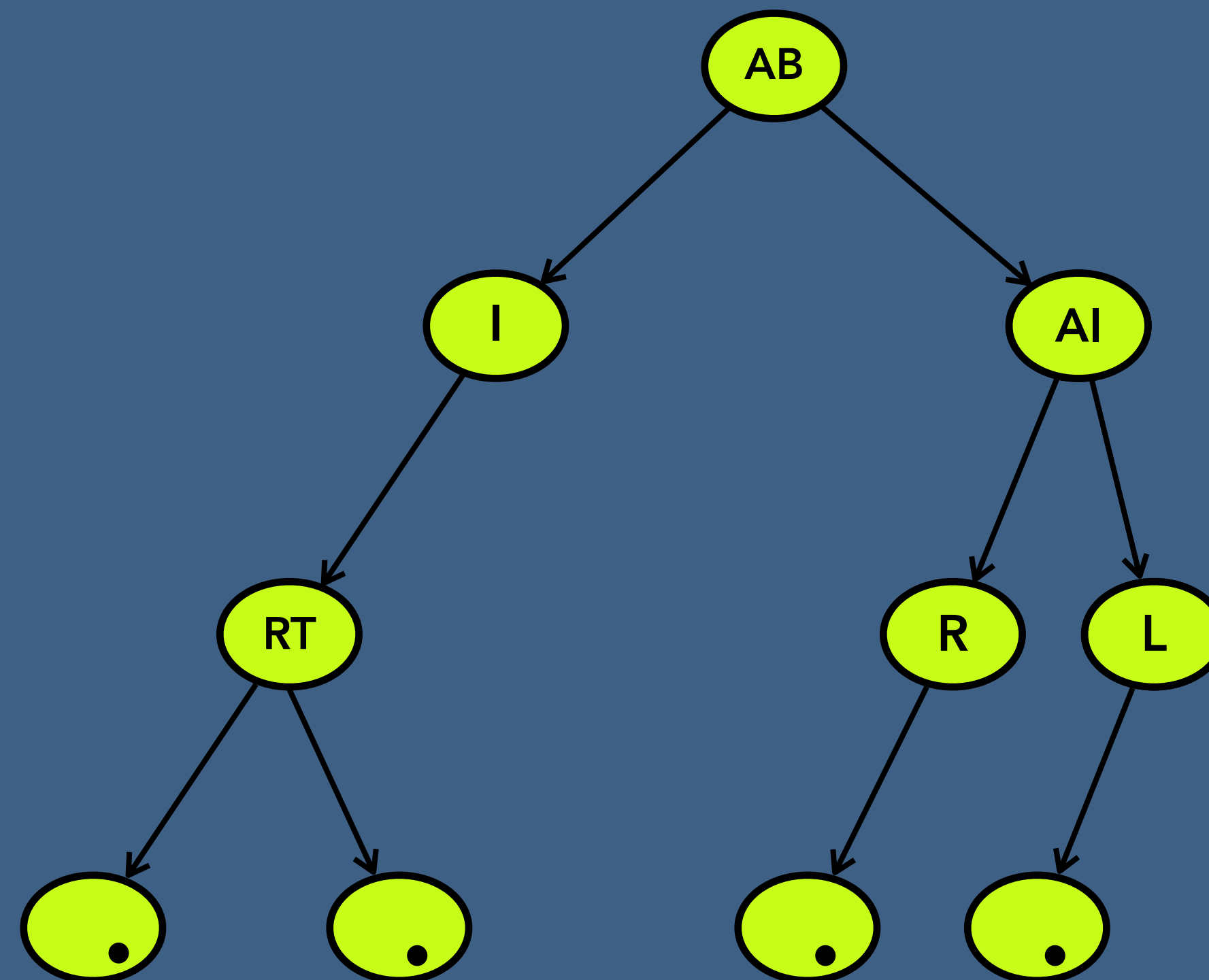
BAR



What is a Trie?

A Trie is a tree-based data structure that organizes information in a hierarchy.

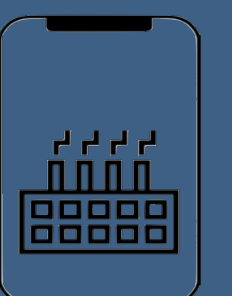
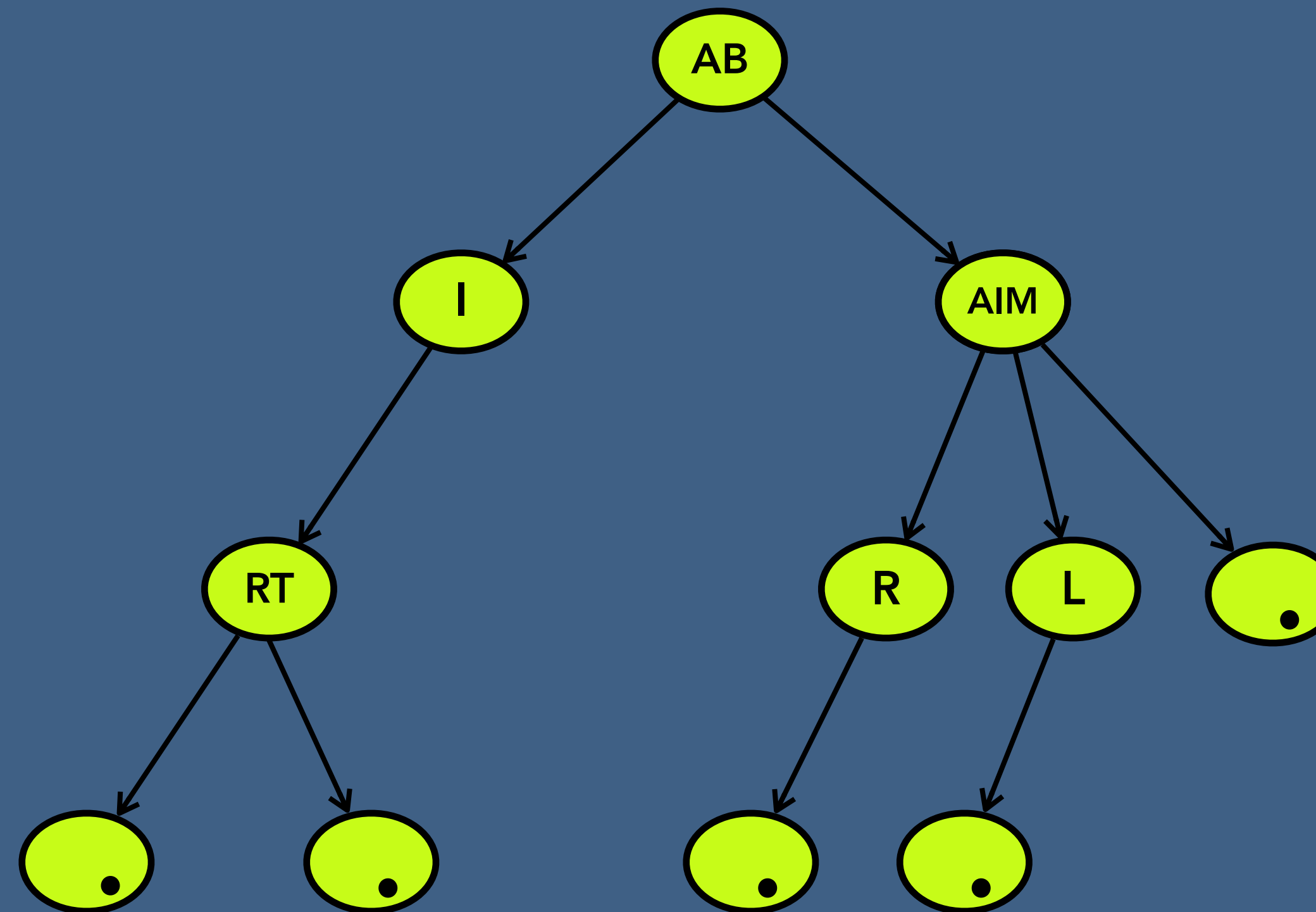
BIL



What is a Trie?

A Trie is a tree-based data structure that organizes information in a hierarchy.

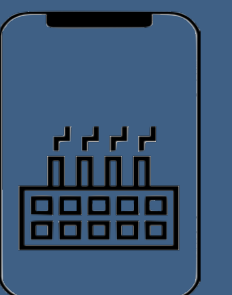
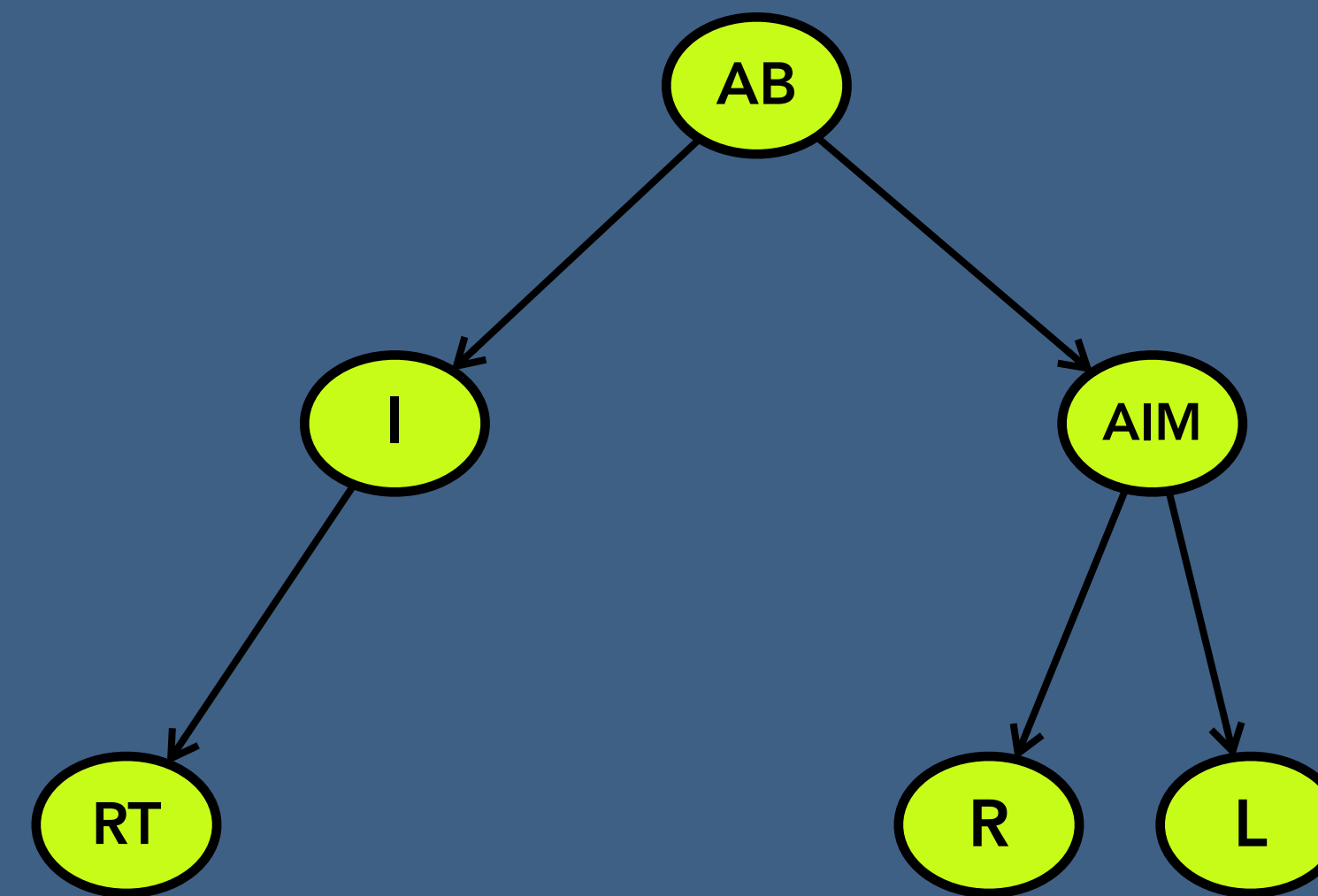
BM



When we need Trie?

To solve many standard problems in efficient way

- Spelling checker
- Auto completion

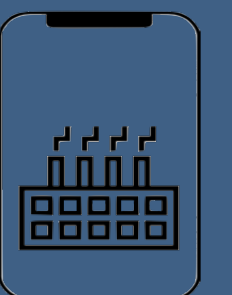
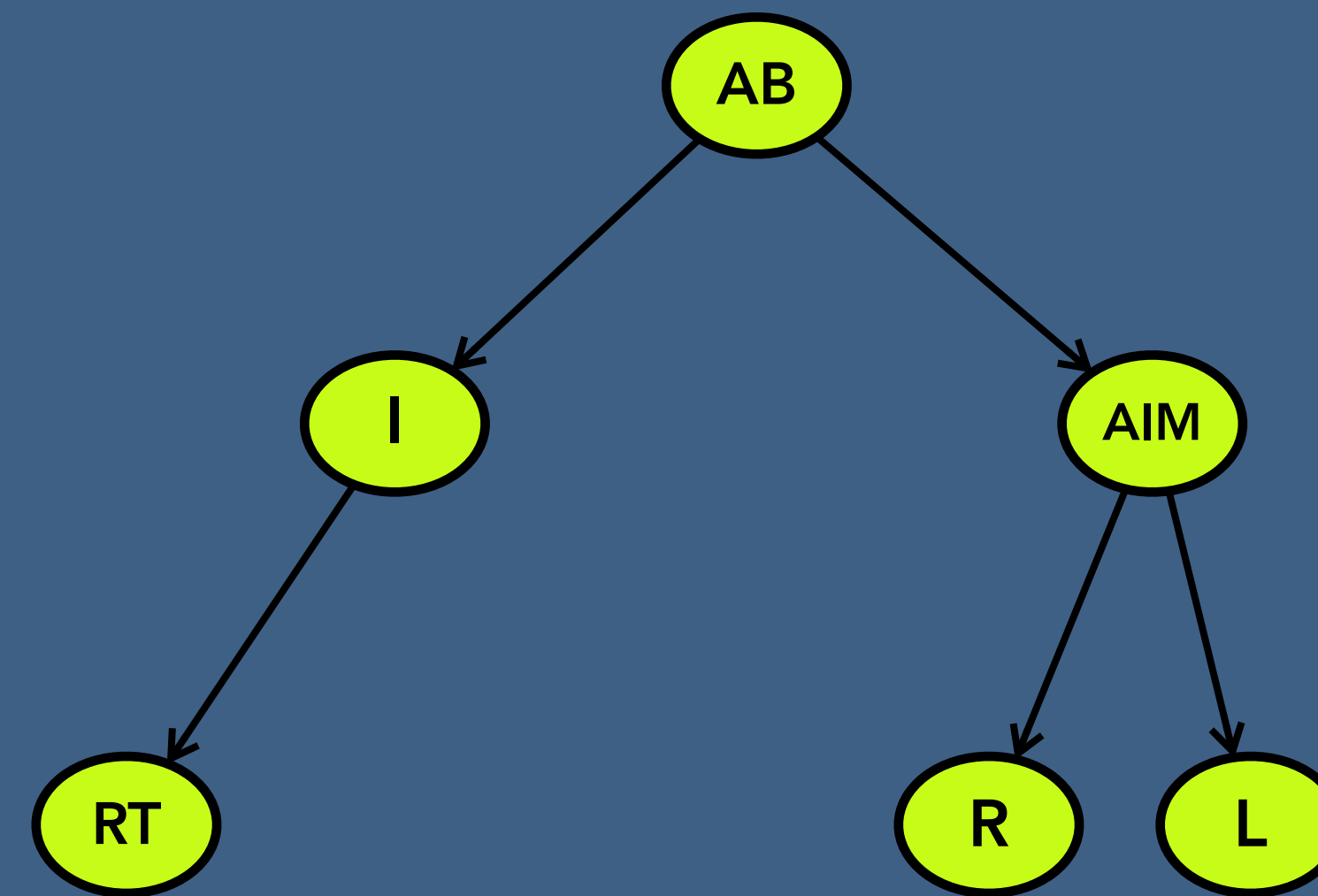


Why we need Trie?

To solve many standard problems in efficient way

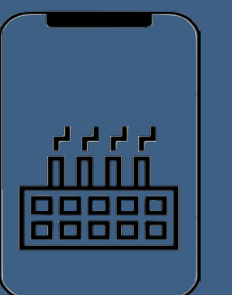
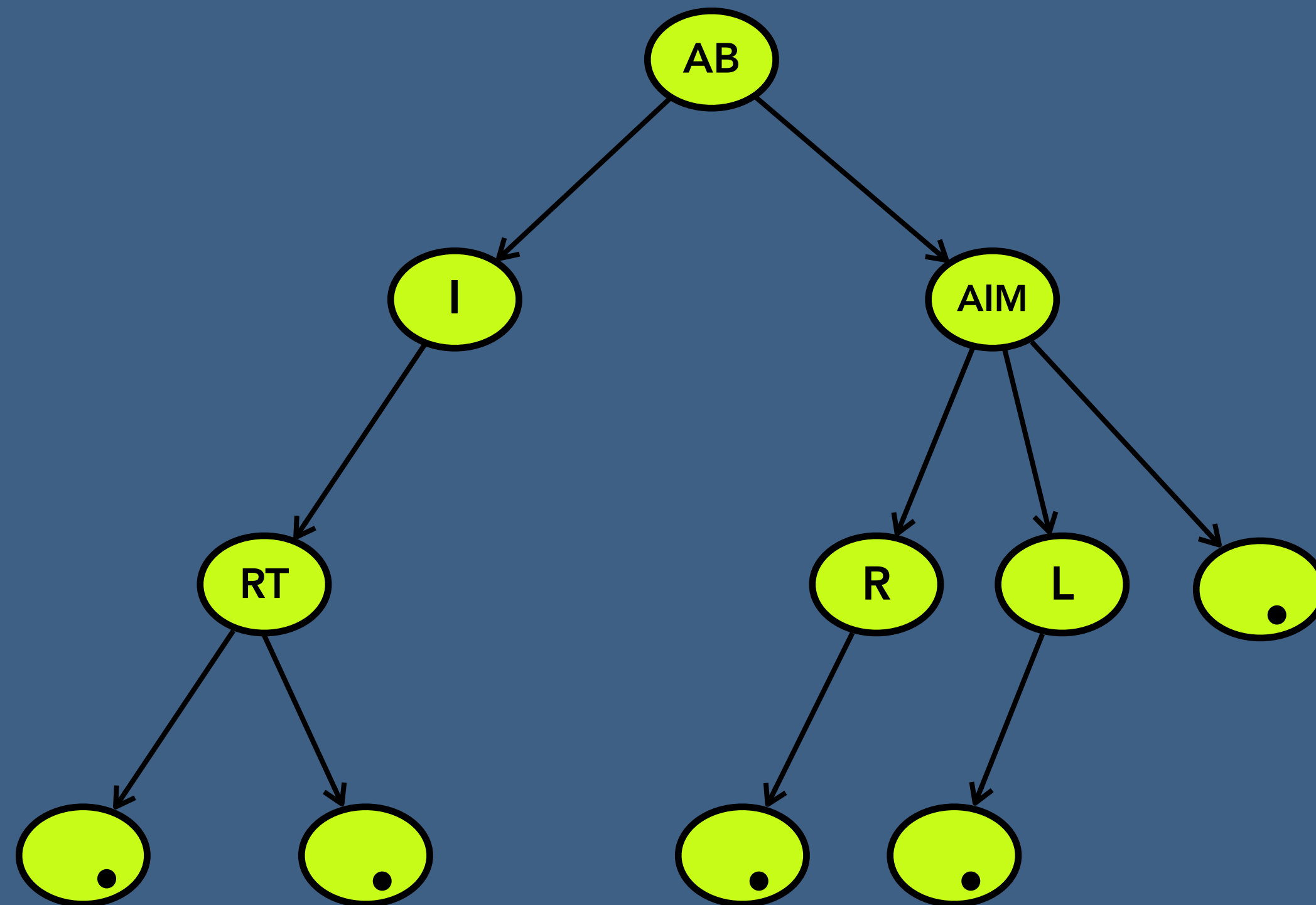
- Spelling checker
- Auto completion

Map	
Characters	Link to Trie Node
End of String	



Common Operations on Trie

- Creation of Trie
- Insertion in Trie
- Search for a String in trie
- Deletion from Trie



Common Operations on Trie

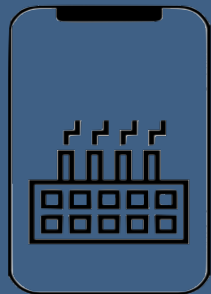
- Creation of Trie



Initialize Trie() class

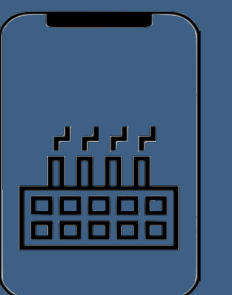
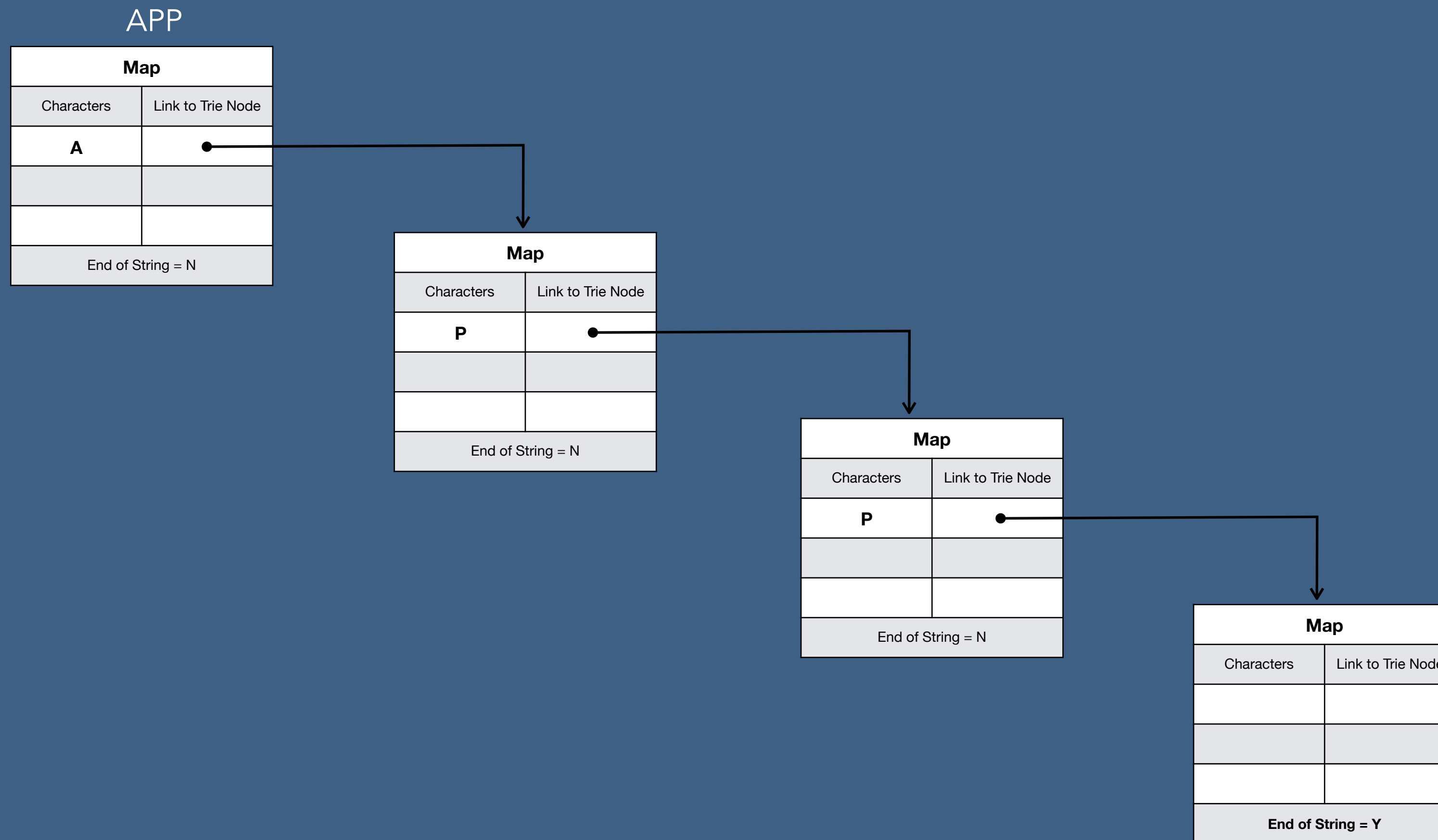
Physically

Map	
Characters	Link to Trie Node
End of String	



Insert a String in a Trie

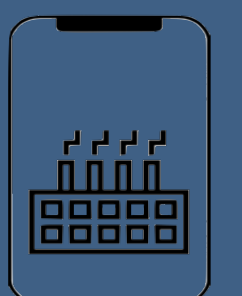
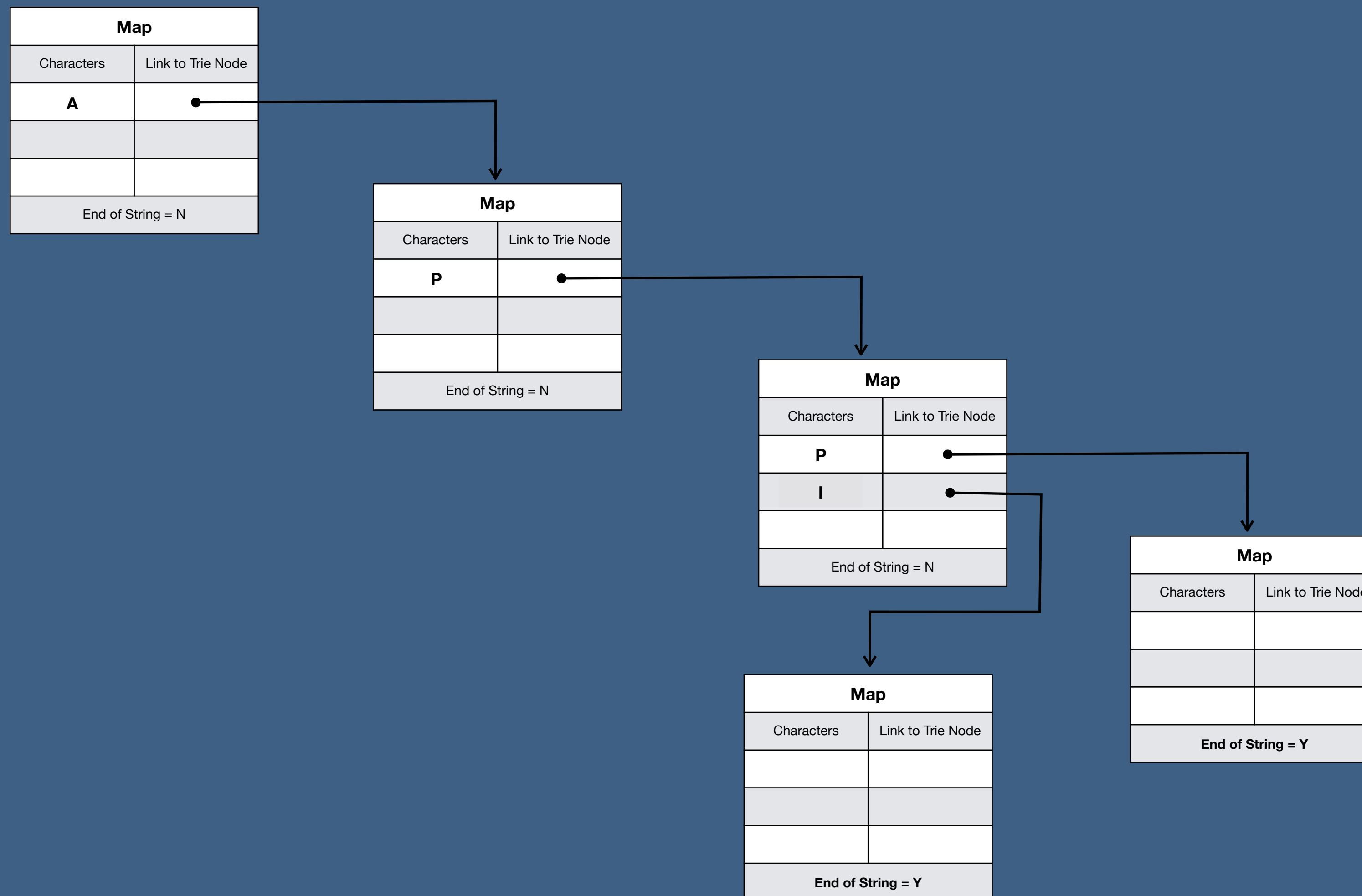
Case 1 : A Trie is Blank



Insert a String in a Trie

Case 2: New string's prefix is common to another strings prefix

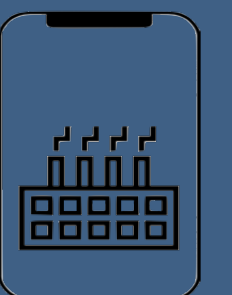
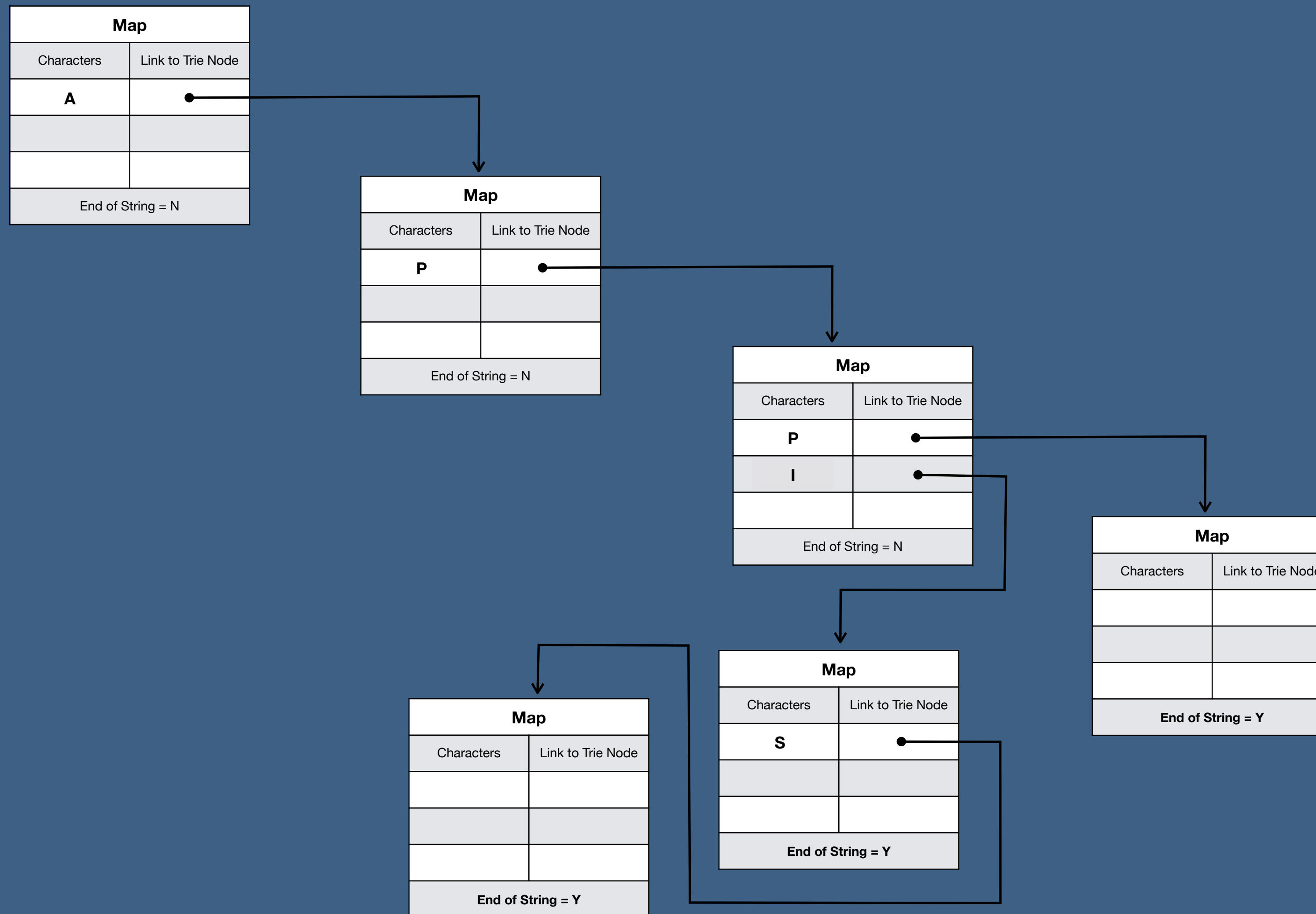
API



Insert a String in a Trie

Case 3: New string's prefix is already present as complete string

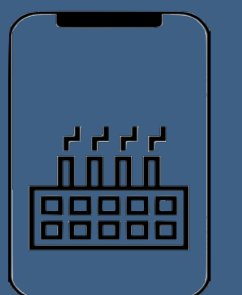
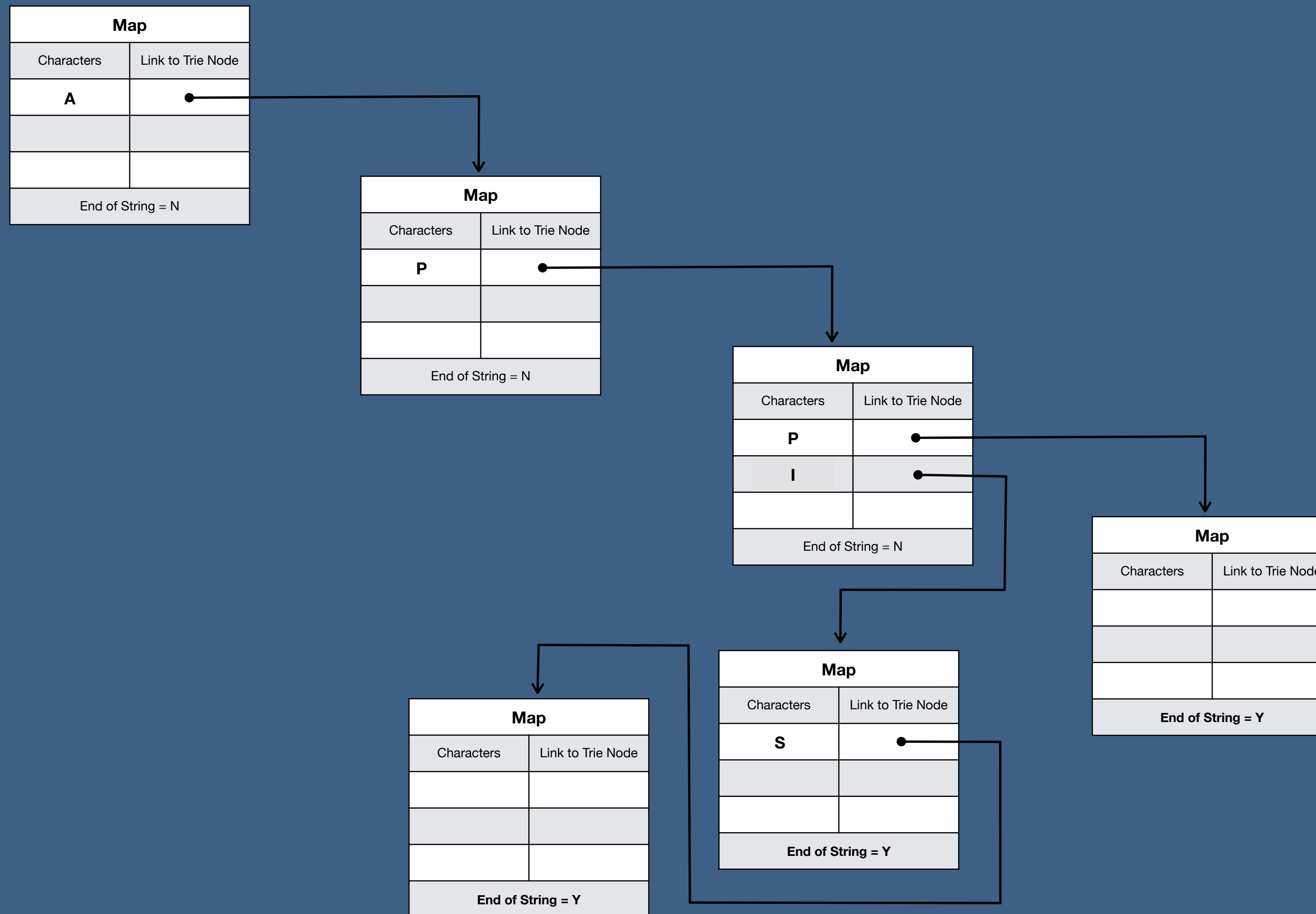
APIS



Insert a String in a Trie

Case 4: String to be inserted is already presented in Trie

APIS

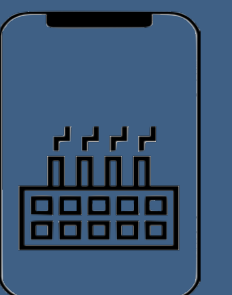
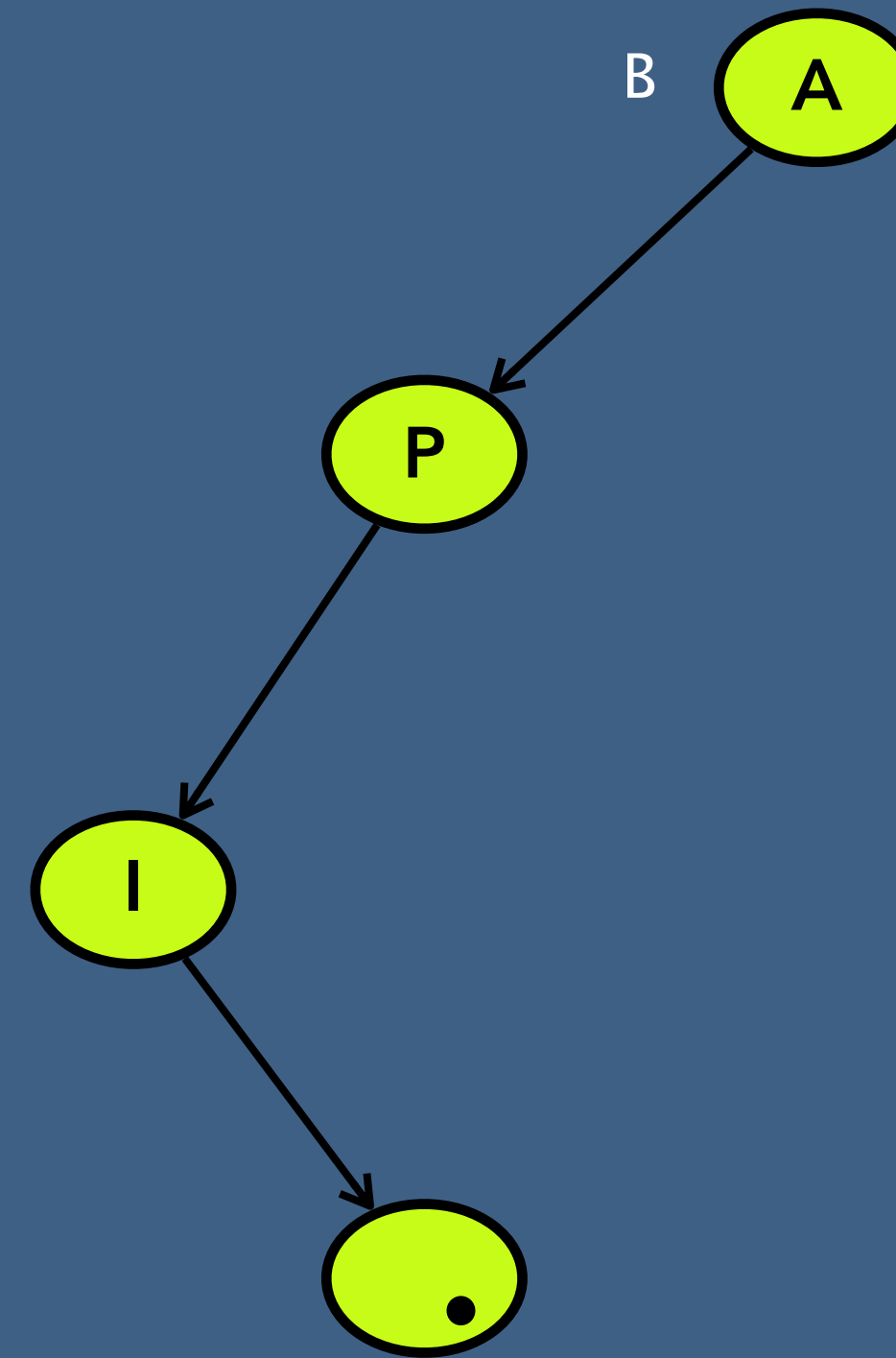


Search for a String in a Trie

Case 1: String does not exist in Trie

BCD

Return : The string does not exist in Trie

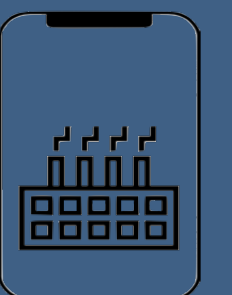
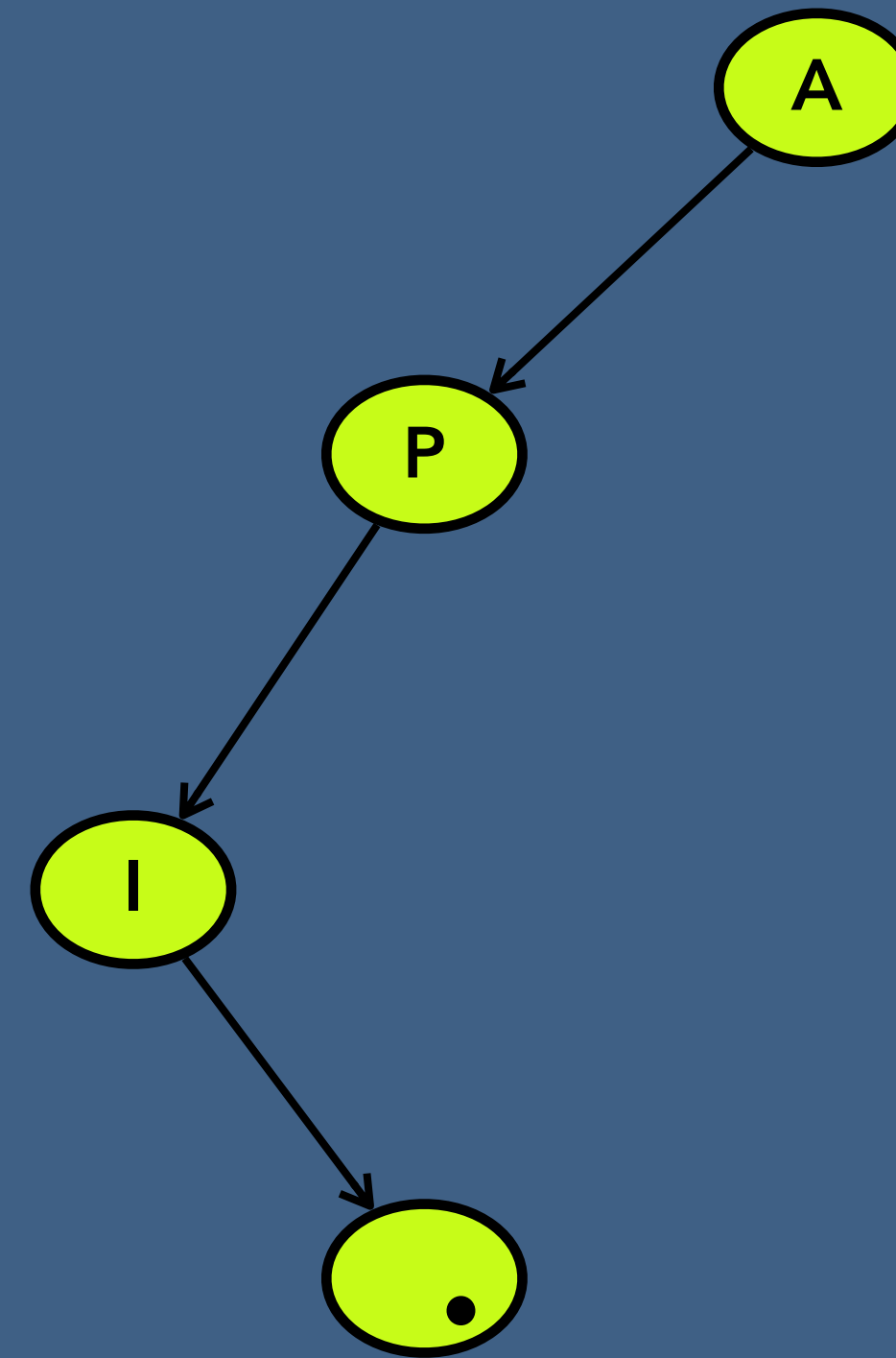


Search for a String in a Trie

Case 2: String exists in Trie

API

Return : TRUE

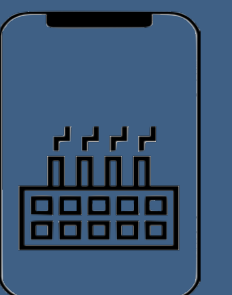
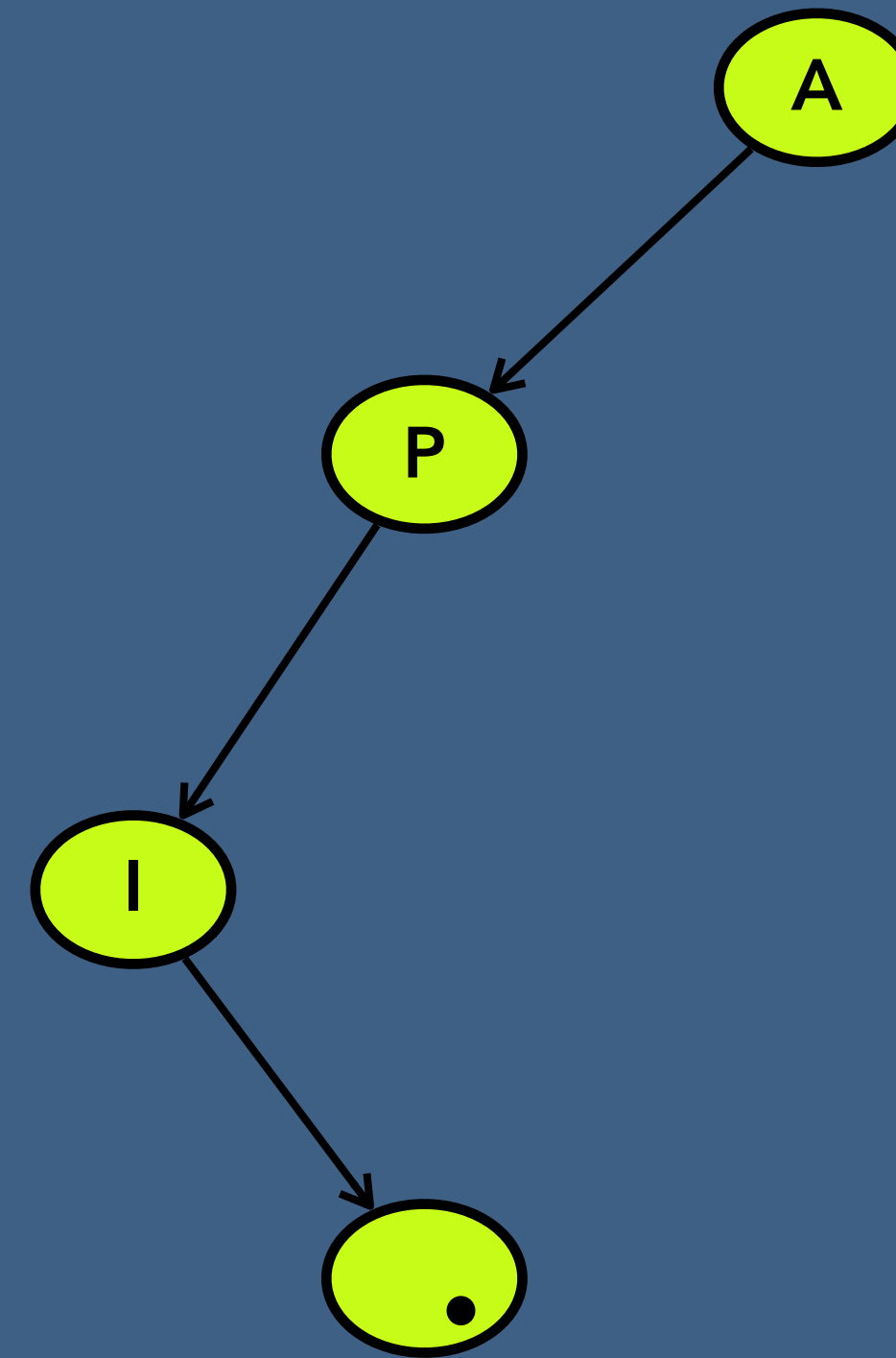


Search for a String in a Trie

Case 3: String is a prefix of another string, but it does not exist in a Trie

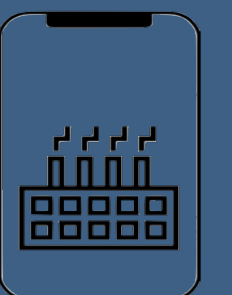
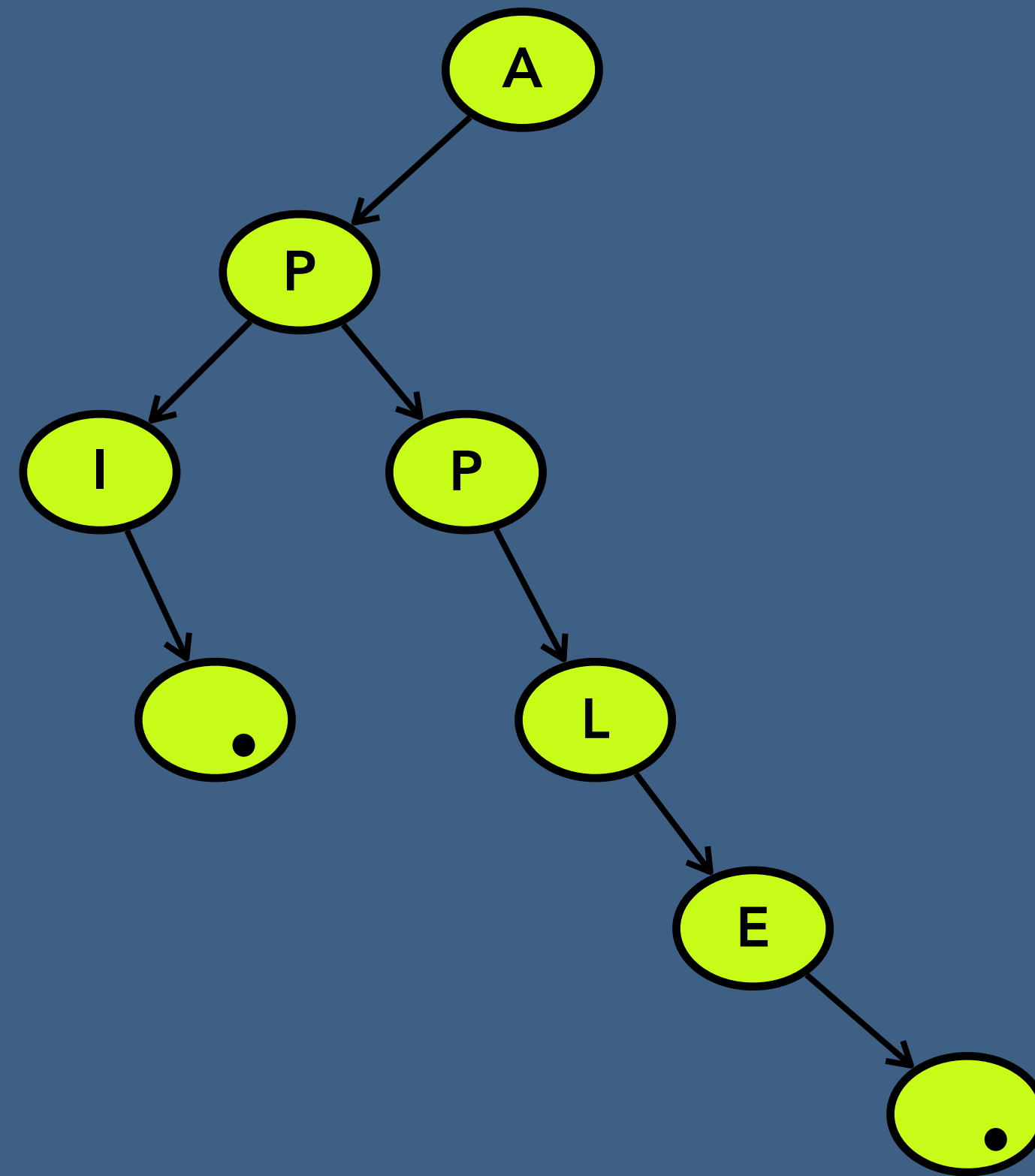
AP

Return : FALSE



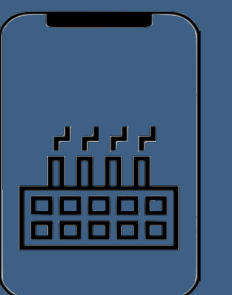
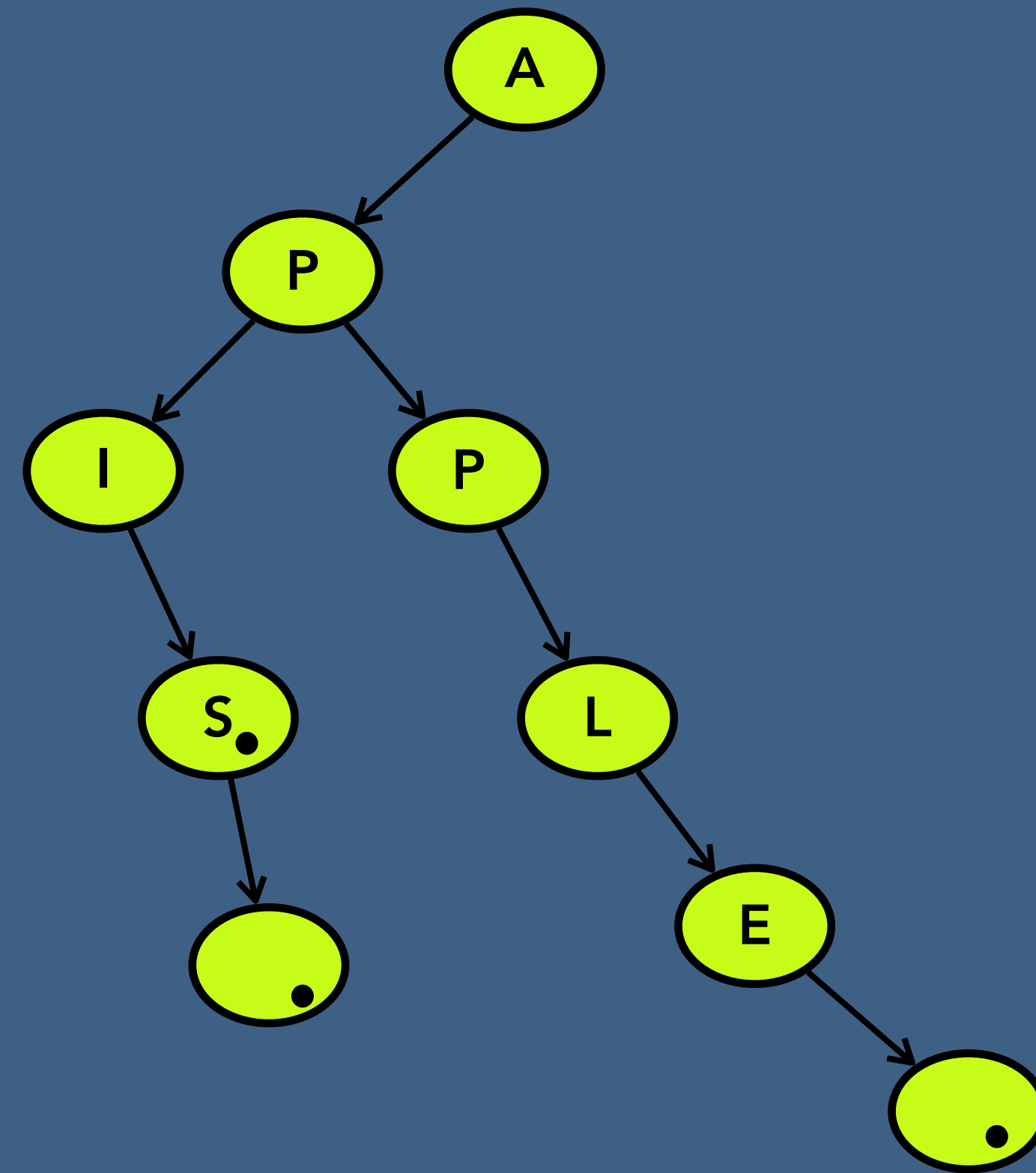
Delete a String from Trie

Case 1: Some other prefix of string is same as the one that we want to delete. (API, APPLE)



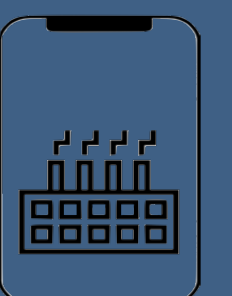
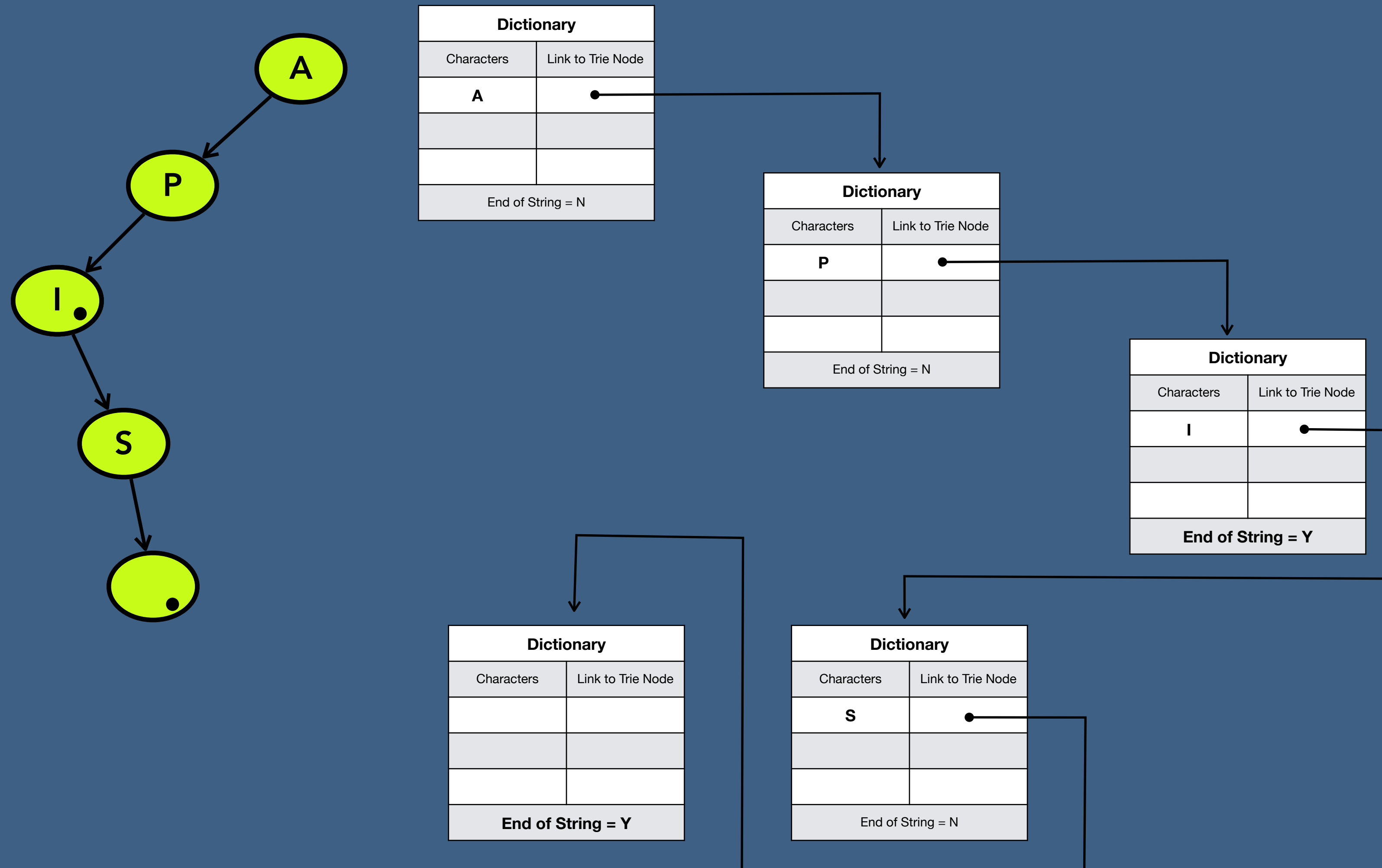
Delete a String from Trie

Case 2: The string is a prefix of another string. (API, APIS)



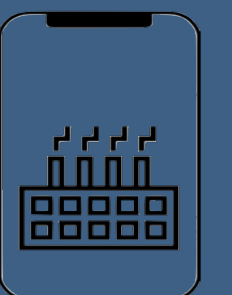
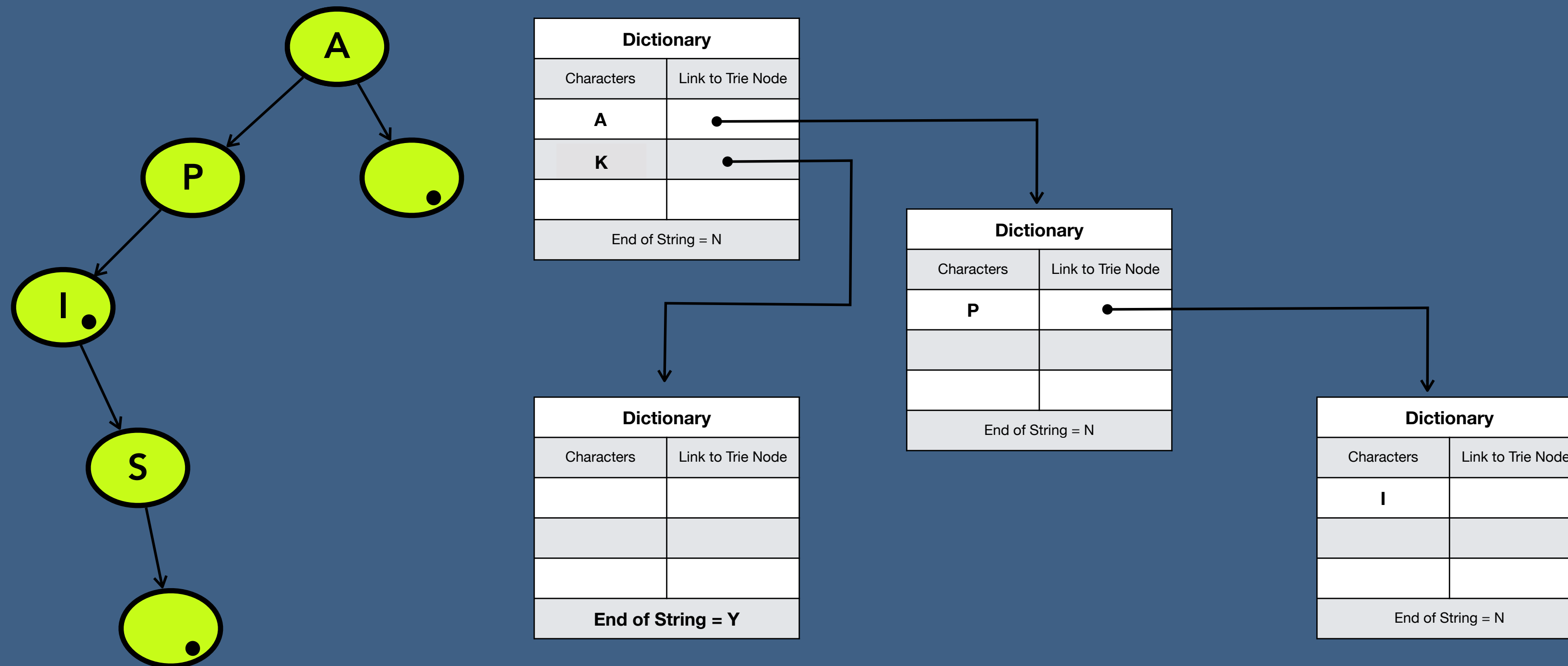
Delete a String from Trie

Case 3: Other string is a prefix of this string. (APIS, AP)



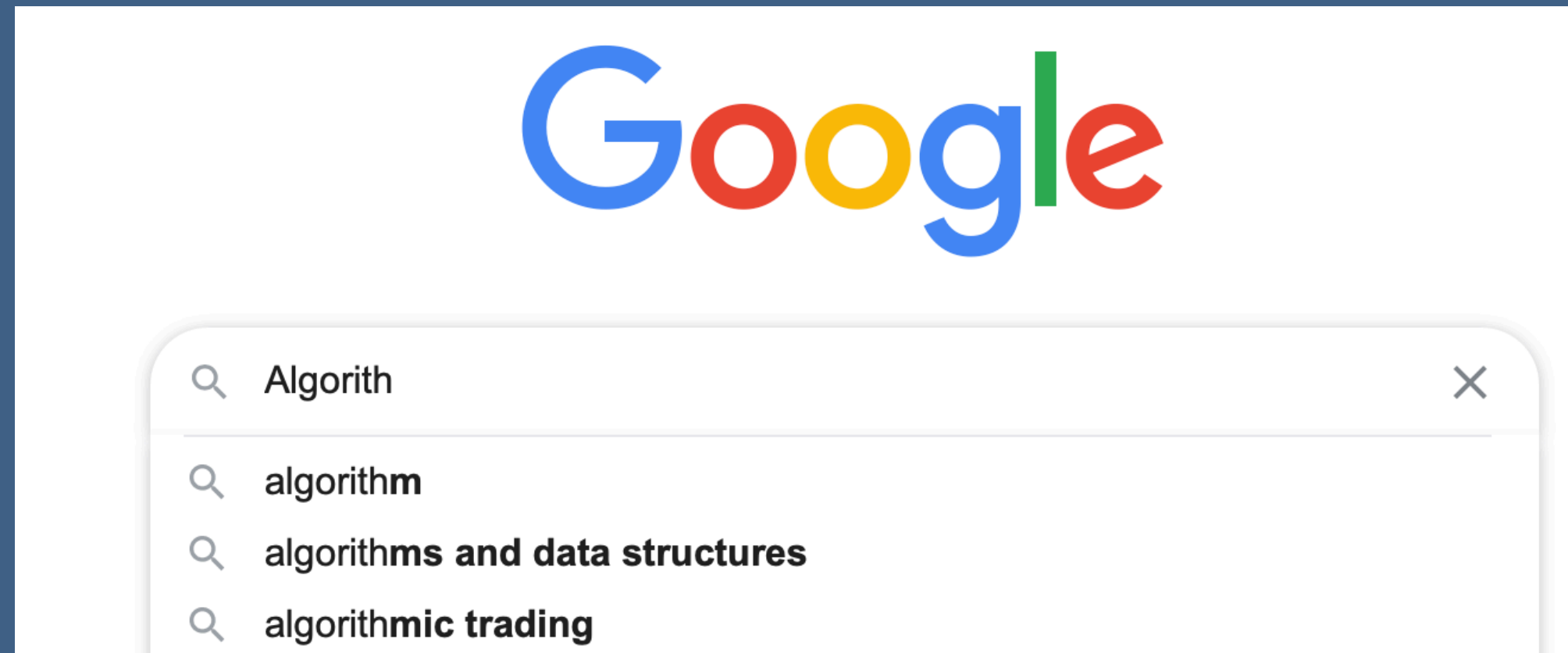
Delete a String from Trie

Case 4: Not any node depends on this String (K)



Practical use of Trie

- Auto completion



- Spelling checker

